



Application Descriptions

7

Energy Management

80

eMobility

3

Summary

This document specifies the Functional Blocks for the control of eMobility appliances and energy management.

This document is WGI Approved.

Document updates

Version	Date	Modifications
1.0	2024_07_19	First version
1.1	2024_11_21	Comments of WGI adjusted and reduced to one function block
1.2	2024_12_16	Complete overhaul, added function block profiles as well as the failsafe chapter and new description of failsafe datapoints. Deleted Connector Error Code since it is already covered in ErrorCode DP
1.3	2025_01_09	Changed description of Failsafe Timeout. Entries in the function block profile table with no content where changed from “-“ to “n.a.”. Added ranges to the datapoints. Unit of EV Battery Capacity changed from [W] to [Wh] Changed the format of EV Departure time from a countdown to a time and date value. Added exception handling for Min/Max total active power, that the values of min can't exceed the max values and vice versa. Changed Target Energy request from [W] to [Wh] Deleted Failsafe Timeout in Connector since the Datapoint is also covered in the Station.
1.4		
01.01.01	2025.08.03	Creation of the WGI Approved version.

References

None.

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Contents

2	Application e-mobility	4
2.1	Introduction.....	4
2.2	Functional Block: EVSE AC	4
2.2.1	Introduction.....	4
2.2.2	Failsafe Mode.....	5
2.2.3	Function Block Diagram: EVSE AC	5
2.2.4	List of Datapoints: EVSE AC	6
2.2.5	Function Block Profiles	9
2.2.6	Inputs, Outputs, Parameters and Diagnostic Data of EVSE AC Station	11
2.2.7	Inputs, Outputs, Parameters and Diagnostic Data of EVSE AC Connector.....	30
2.2.8	New Data Point Types	55

2 Application e-mobility

2.1 Introduction

E-mobility refers to the use of electric vehicles (EVs) powered by electricity instead of traditional fuels. A key component of e-mobility is the network of charging stations, or “charging points”, which are essentially for recharging EV batteries. These stations range from home chargers to public fast chargers with one or more charging points for EVs depending on the model. Therefore, charging station and charging point are split. This helps to better include the full e-mobility functionality inside the energy management.

EVSE stands for Electric Vehicle Supply Equipment. It's the infrastructure used to charge electric vehicles. EVSE typically includes charging stations, cables and connectors. It is used to charge vehicles at home, workplaces, at public places and other locations. It provides the necessary electrical connection and safety features to charge EVs safely and efficiently.

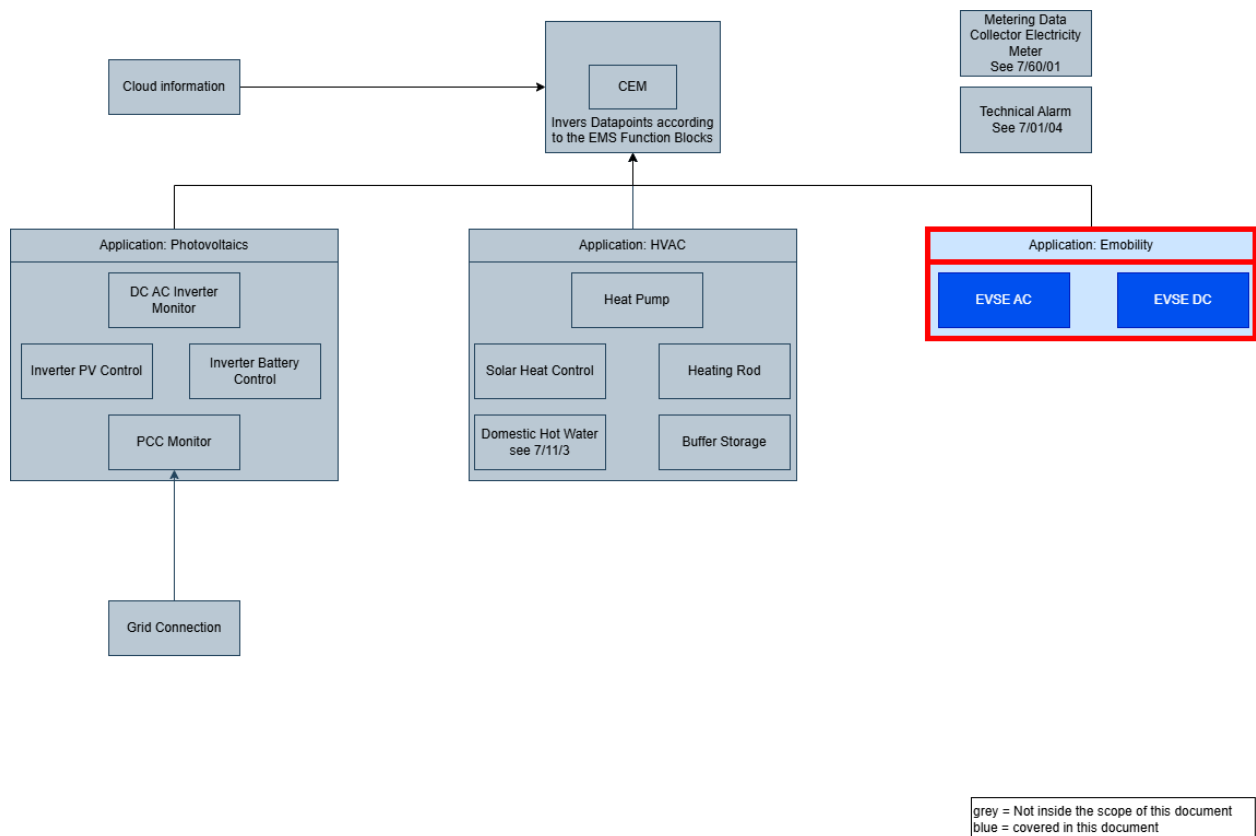


Figure 1 - Application “eMobility”

2.2 Functional Block: EVSE AC

2.2.1 Introduction

The function block EVSE AC lays down the basic functions of an electric vehicle AC charging station, to optimize the charging pattern and react to possible curtailment issues. Therefore, the properties of this functional block are subdivided in charging station and connector. If a charging station consist of multiple charging points, the properties and datapoints related to the connector can be multiplied. This provides a high degree of energy management flexibility, especially for larger charging systems.

The two existing types of charging stations are based on AC or DC: Typically, the power output of an AC station is set up through a Current limit. The DC Station normally uses a Power Limit. Therefore, the function block realises the difference through the two Datapoints EVSE Current Limit Set and EVSE Total Active Power Limit Set, which are both optional. In this way the manufacturer can implement AC or DC stations by choosing the Datapoint best suited for his device

2.2.2 Failsafe Mode

Failsafe mode describes the state in which the device loses the connection to the HEMS. If a signal is not detected at any of the inputs of the function block within the set time in the FailSafeTimeout data point, it is assumed that the connection to the HEMS has been lost. As a result, the fail-safe values for current or power are set as new limit values in the device. The FailSafeFeedback data points are intended for visualization purposes to indicate a loss of communication with the HEMS to the KNX installation.

2.2.3 Function Block Diagram: EVSE AC

EVSE			
Inputs	Parameter	Description	Output
CurrentLimit	CurrentLimit		
Active Power Limit	Active Power Limit		
PhaseSwitching	PhaseSwitching		
		CurrentLimitEff	CurrentLimitEff
		Failsafe	Failsafe
		ActivePowerLimitEff	ActivePowerLimitEff
		FailsafeCurrentEff	FailsafeCurrentEff
		FailsafeActivePowerEff	FailsafeActivePowerEff
		ActivePower	ActivePower
		CurrentL1	CurrentL1
		CurrentL2	CurrentL2
		CurrentL3	CurrentL3
		TempPCBA	TempPCBA
		ErrorCode	ErrorCode
		PhaseSwitchingEff	PhaseSwitchingEff
	FailSafeActivePower		
	FailSafeCurrent		
	FailSafeTimeout		
		ActivePowerMax	
		ActivePowerMin	
		CurrentRated	
		CurrentMax	
		CurrentMin	
CurrentLimit	CurrentLimit		
ActivePowerLimit	ActivePowerLimit		
		CurrentLimitEff	CurrentLimitEff
		FailsafeCurrentEff	FailsafeCurrentEff
		ActivePowerLimitEff	ActivePowerLimitEff
		FailsafeActivePowerEff	FailsafeActivePowerEff
		ActivePower	ActivePower
		CurrentL1	CurrentL1
		CurrentL2	CurrentL2
		CurrentL3	CurrentL3
		ControlPilotState	ControlPilotState
		OCPPState	OCPPState
		ISO15118Detected	ISO15118Detected
		StateOfCharge	StateOfCharge
		BatteryCapacity	BatteryCapacity
		DepartureTime	DepartureTime
		StateOfChargeMin	StateOfChargeMin
		StateOfChargeTarget	StateOfChargeTarget
		StateOfChargeTargetEnergyReq	StateOfChargeTargetEnergyReq
		RFIDTag	RFIDTag
	FailSafeActivePower		
	FailSafeCurrent		
		ActivePowerMax	
		ActivePowerMin	
		CurrentMax	
		CurrentMin	
		BidirectionalEVDetected	
		DischargePowerMax	
		DischargePowerMin	

Figure 2: EVSE

2.2.4 List of Datapoints: EVSE AC

2.2.4.1 General

Inputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentLimit	Input for the upper limit of the current of the charging point that can be supplied to the vehicle. 0A pauses the charging session.	14.019 DPT_Value_Electric_Current
Active Power Limit	Input for the upper limit of the power of the charging point that can be supplied to the vehicle. 0kW pauses the charging session.	14.056 DPT_Value_Power
PhaseSwitching	This input enables the switching between 3 to 1 phase or the other way around. This function is required for PV surplus charging. 0 - activate L1, L2, L3; 1 - activate only L1	1.001 DPT_Switch

Outputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentLimitEff	Feedback for the upper limit of the current of the charging point that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current
Failsafe	Feedback when the charging station has switched to fail-safe mode	1.002 DPT_Bool
ActivePowerLimitEff	Feedback for the upper limit of the power of the charging point that can be supplied to the vehicle	14.056 DPT_Value_Power
FailsafeCurrentEff	Feedback of the current used during failsafe mode in station	14.019 DPT_Value_Electric_Current
FailsafeActivePowerEff	Feedback of the power used during failsafe mode in station	14.056 DPT_Value_Power
ActivePower	Present power of the charging point	14.056 DPT_Value_Power
CurrentL1	Present Current of the charging point on phase 1	14.019 DPT_Value_Electric_Current
CurrentL2	Present Current of the charging point on phase 2	14.019 DPT_Value_Electric_Current
CurrentL3	Present Current of the charging point on phase 3	14.019 DPT_Value_Electric_Current
TempPCBA	Present internal temperature of the charging point	9.001 DPT_Value_Temp
ErrorCode	error code of the charging point	12.1221 DPT_EVSEMode_3
PhaseSwitchingEff	This output shows the number of phases used for charging. This function is required for PV surplus charging. 0 - activate L1, L2, L3; 1 - activate only L1	1.001 DPT_Switch

Parameters		
Datapoint	Description/Remarks	Datapoint Type
FailSafeActivePower	Maximum power that the station may draw when it is in failsafe mode	14.056 DPT_Value_Power
FailSafeCurrent	Maximum current that the station may draw when it is in failsafe mode	14.019 DPT_Value_Electric_Current
FailsafeTimeout	The time value that is used to detect a failure. If no value on the inputs is received during this time, the failsafe values will be activated and the failsafe feedback outputs will be set	7.005 DPT_TimePeriodSec

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
ActivePowerMax	Maximum power of the charging point that can be supplied to the vehicle	14.056 DPT_Value_Power
ActivePowerMin	Minimum power of the charging point that can be supplied to the vehicle	14.056 DPT_Value_Power
CurrentRated	Rated current of the charging point	14.019 DPT_Value_Electric_Current
CurrentMax	Maximum current of the charging point that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current
CurrentMin	Minimum current of the charging point, that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current

2.2.4.2 Connector Datapoints

Inputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentLimit	Input for the upper limit of the current of the charging point that can be supplied to the vehicle. 0A pauses the charging session.	14.019 DPT_Value_Electric_Current
ActivePowerLimitEff	Input for the upper limit of the power of the charging point that can be supplied to the vehicle. 0kW pauses the charging session.	14.056 DPT_Value_Power

Outputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentLimitEff	Feedback for the upper limit of the current of the charging point that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current
FailsafeCurrentEff	Feedback of the current used during failsafe mode in a connector	14.019 DPT_Value_Electric_Current
ActivePowerLimitEff	Feedback for the upper limit of the power of the charging point that can be supplied to the vehicle	14.056 DPT_Value_Power
FailsafeActivePowerEff	Feedback of the power used during failsafe mode in station	14.056 DPT_Value_Power
ActivePower	Present power of the charging point	14.056 DPT_Value_Power
CurrentL1	Present Current of the charging point on phase 1	14.019 DPT_Value_Electric_Current

Outputs		
Datapoint	Description/Remarks	Datapoint Type
CurrentL2	Present Current of the charging point on phase 2	14.019 DPT_Value_Electric_Current
CurrentL3	Present Current of the charging point on phase 3	14.019 DPT_Value_Electric_Current
ControlPilotState	Enum for different states of the chargingpoint	20.1219 DPT_EVSEMode_1
OCPPState	States of the Charging Point according to OCPP	20.1220 DPT_EVSEMode_2
ISO15118Detected	High level (digital) communication with the car is possible	1.002 DPT_Bool
StateOfCharge	State of charge of electric vehicle in %	5.001 DPT_Scaling
BatteryCapacity	Battery capacity of electric vehicle in Watt	14.056 DPT_Value_Power
DepartureTime	Departure time of electric vehicle in seconds since start of charging session	7.005 DPT_TimePeriodSec
StateOfChargeMin	Minimum State of Charge the EV needs to keep throughout the charging session	5.001 DPT_Scaling
StateOfChargeTarget	Target SoC for departure	5.001 DPT_Scaling
StateOfChargeTargetEnergyReq	Energy required to reach target SoC for departure	13.010 DPT_ActiveEnergy
RFIDTag	RFID of the plugged in EV	24.001 DPT_VarString_8859_1

Parameters		
Datapoint	Description/Remarks	Datapoint Type
FailSafeActivePower	Maximum power that the station may draw when it is in failsafe mode	14.056 DPT_Value_Power
FailSafeCurrent	Maximum current that the station may draw when it is in failsafe mode	14.019 DPT_Value_Electric_Current

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
ActivePowerMax	maximum power of the charging point that can be supplied to the vehicle	14.056 DPT_Value_Power
ActivePowerMin	minimum power of the charging point, that can be supplied to the vehicle	14.056 DPT_Value_Power
CurrentMax	maximum current of the charging point that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current
CurrentMin	minimum current of the charging point, that can be supplied to the vehicle	14.019 DPT_Value_Electric_Current
BidirectionalEVDetected	Indicates whether bidirectional charging is supported by the charging station/EV	1.002 DPT_Bool
DischargePowerMax	Max possible power that can be drawn from the EV through bidirectional charging	14.056 DPT_Value_Power
DischargePowerMin	Min possible power that can be drawn from the EV through bidirectional charging	14.056 DPT_Value_Power

2.2.5 Function Block Profiles

The following profiles can be used to create different types of charging stations.

- Profile 1 Current Control:
This profile can be used to create charging stations that are controlled by current
- Profile 2 Power Control:
This profile can be used to create charging stations that are controlled by power

Profiles 3 and 4 can be used to create additional charging points if the station has more than one charging point. Profile 3 is used for current controlled stations, profile 4 is used for power controlled stations.

Datapoint	Profile 1 Current control	Profile 2 Power Control	Profile 3 Additional current controlled connectors	Profile 4 Additional power controlled connectors
Station				
CurrentLimit	M	O	NA	NA
Active Power Limit	O	M	NA	NA
PhaseSwitching	O	O	NA	NA
CurrentLimitEff	M	O	NA	NA
Failsafe	M	M	NA	NA
ActivePowerLimitEff	O	M	NA	NA
FailsafeCurrentEff	M	O	NA	NA
FailsafeActivePowerEff	O	M	NA	NA
ActivePower	O	M	NA	NA
CurrentL1	M	O	NA	NA
CurrentL2	M	O	NA	NA
CurrentL3	M	O	NA	NA
TempPCBA	O	O	NA	NA
ErrorCode	M	M	NA	NA
PhaseSwitchingEff	M	M	NA	NA
FailSafeActivePower	O	M	NA	NA
FailSafeCurrent	M	O	NA	NA
FailsafeTimeout	M	M	NA	NA
ActivePowerMax	O	M	NA	NA
ActivePowerMin	O	M	NA	NA
CurrentRated	M	O	NA	NA
CurrentMax	M	O	NA	NA
CurrentMin	M	O	NA	NA
Connector				
CurrentLimit	M	O	M	O
ActivePowerLimitEff	O	M	O	M

CurrentLimitEff	M	O	M	O
FailsafeCurrentEff	M	O	M	O
ActivePowerLimitEff	O	M	O	M
FailsafeActivePowerEff	O	M	O	M
ActivePower	O	M	O	M
CurrentL1	M	O	M	O
CurrentL2	M	O	M	O
CurrentL3	M	O	M	O
ControlPilotState	O	O	O	O
OCPState	O	O	O	O
ISO15118Detected	M	M	M	M
StateOfCharge	O	O	O	O
BatteryCapacity	O	O	O	O
DepartureTime	O	O	O	O
StateOfChargeMin	O	O	O	O
StateOfChargeTarget	O	O	O	O
StateOfChargeTargetEnergyReq	O	O	O	O
RFIDTag	O	O	O	O
FailSafeActivePower	O	M	O	M
FailSafeCurrent	M	O	M	O
ActivePowerMax	O	M	O	M
ActivePowerMin	O	M	O	M
CurrentMax	M	O	M	O
CurrentMin	M	O	M	O
BidirectionalEVDetected	M	M	M	M
DischargePowerMax	M	M	M	M
DischargePowerMin	M	M	M	M

2.2.6 Inputs, Outputs, Parameters and Diagnostic Data of EVSE AC Station

2.2.6.1 Inputs: EVSE AC Station

2.2.6.1.1 CurrentLimit

DP Name:	CurrentLimit	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Input for the upper limit of the current of the charging point that can be supplied to the vehicle. 0 Ampere pauses the charging session					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float Value	Setting the upper limit of current of charging point to vehicle	M	0-10.000	A	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Values below the minimal charging current of the station will automatically set the value to 0 Ampere. Values above the maximal charging current will set the value to maximal charging current. For FailSafe, see FailsafeTimeout					
Special Features					

2.2.6.1.2 ActivePowerLimit

DP Name:	ActivePowerLimit	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Input for the upper limit of the power of the charging point which can be supplied to the vehicle. 0kW pauses the charging session.					
Datapoint Type					
DPT_Name:	DPT_Value_Power				
DPT Format:		DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float Value	Setting the upper limit of the charging point power	M	0-10.000.000	W	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	52	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Values below the minimal charging power of the station will automatically set the value to 0 kW. Adding values above the maximal charging power will set the value to Max Power. For FailSafe, see FailSafeTimeout					
Special Features					

2.2.6.1.3 PhaseSwitching

DP Name:	PhaseSwitching		Abbr.:		Mandatory	☐
FB Name:	EVSE				Can be internal	☐
Description						
This input enables the switching between 3 to 1 phase or the other way around. This function is required for PV surplus charging. 0 - activate L1, L2, L3 (0 = off) 1 - activate only L1 (1 = on)						
Datapoint Type						
DPT Name:						
DPT Format:	DPT Switch		DPT ID:	1.001		
Field			Supp.	Range	Unit	Default
b	Setting 3 to 1 phase or the other way		M	full	bool	none
Access Type						
◆ Input						
N → this	☒	1 → this	☐			
Spontaneous	☒	Cyclically:	☐	Time-out:		
Request	☐	Polling:	☐	Period:		
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	53		
	Start-Index:	1	N° of elements	1		
Property access:	Read only	☐	Read/Write	☒		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

2.2.6.2 Outputs: EVSE AC Station

2.2.6.2.1 CurrentLimitEff

DP Name:	CurrentLimitEff	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Feedback for the upper limit of the current of the charging point that can be supplied to the vehicle					
Datapoint Type					
DPT Name:	DPT Value Electric Current				
DPT Format:		DPT ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing the upper limit of the current of the charging point	M	0-10.000	A	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	70	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.6.2.2 FailSafe

DP Name:	FailSafe	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Feedback when the charging station has switched to fail-safe mode					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Bool	DPT_ID:	1.002		
Field		Supp.	Range	Unit	Default
b	Value informing that charging station has switched to fail-safe mode (1 = failsafe active, 0 = failsafe not active)	M	full	bool	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	71
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.6.2.3 ActivePowerLimitEff

DP Name:	ActivePowerLimitEff	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Feedback for the upper limit of the power of the charging point that can be supplied to the vehicle					
Datapoint Type					
DPT Name:					
DPT Format:	DPT Value Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value representing the upper power limit of the charging point	M	0-10.000.000	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	ΔValue:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	72	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.6.2.4 FailSafeCurrentEff

DP Name:	FailSafeCurrentEff	Abbr.:		Mandatory	☒
FB Name:				Can be internal	☐
Description					
Feedback of the current used during failsafe mode of station					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value reporting on the current used during failsafe mode of station	M	0-10.000	A	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	ΔValue:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	73	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
For FailSafe, see FailsafeTimeout					
Special Features					

2.2.6.2.5 FailSafeActivePowerEff

DP Name:	FailSafeActivePowerEff	Abbr.:		Mandatory	☒
FB Name:				Can be internal	☐
Description					
Feedback of the power used during failsafe mode of station					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value reporting on the power used during failsafe mode of station		0-10.000.000	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	ΔValue:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	74	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
For FailSafe, see FailsafeTimeout					
Special Features					

2.2.6.2.6 ActivePower

DP Name:	ActivePower	Abbr.:		Mandatory	☒
FB Name:				Can be internal	☐
Description					
Present power of the charging point					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value reporting on present power of charging point	M	0-10.000.000	W	none
Access Type					
◆ Output					
	this → M	☒	this → 1	☐	
	Spontaneous	☒	COV:	☒	ΔValue: >1 %
			Cyclic	☐	Period:
	Request	☒			
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	75
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☒	Read/Write	☐
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.6.2.7 CurrentL1

DP Name:	CurrentL1	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present current of the charging point on phase 1					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing present L1 current	M	0-10.000	A	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	ΔValue:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	76	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.6.2.8 CurrentL2

DP Name:	CurrentL2	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present current of the charging point on phase 2					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing present L2 current		0-10.000	A	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	77	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.6.2.9 CurrentL3

DP Name:	CurrentL3	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present current of the charging point on phase 3					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing present L3 current	M	0-10.000	A	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	ΔValue:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	78	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.6.2.10 TempPCBA

DP Name:	TempPCBA	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Present internal temperature of the charging point					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value representing the present internal temperature of the charging point		-50°C- +200°C	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	79	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.6.2.11 ErrorCode

DP Name:	ErrorCode	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Error code of the charging point					
Datapoint Type					
DPT_Name:					
DPT Format:	20.1221	DPT_ID:	DPT_EVSEMode_3		
Field		Supp.	Range	Unit	Default
Field1	Value reflecting the current charging point error code 1 = Connector Lock Failure 2 = Communication failure with the vehicle 3 = Ground Failure 4 = High Temperature 5 = Error in internal hard- or software component 6 = Local List Conflict 7 = NoError 8-255: reserved	M	full	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	80	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.6.2.12 PhaseSwitchingEff

DP Name:	PhaseSwitchingEff		Abbr.:		Mandatory	☒
FB Name:	EVSE				Can be internal	☐
Description						
This output shows the number of phases used for charging. This function is required for PV surplus charging. 0 – activation of L1, L2, L3 1 – activation of only L1						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Switch		DPT ID:	1.001		
Field			Supp.	Range	Unit	Default
b	Value reflecting the number of phases used for charging		M	Full	bool	none
Access Type						
◆ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☒	Δ-Value:		
		Cyclic	☐	Period:		
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	81		
	Start-Index:	1	N° of elements	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

2.2.6.3 Parameter: EVSE AC Station

2.2.6.3.1 FailSafeActivePower

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐	
Description:								
Maximum power that the station may draw when it is in failsafe mode								
DPT:	Name	DPT Value Power	DPT_ID	14.056	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Float	Maximum station power when in failsafe mode		M	0-10.000.000	W	--	none	
Communication:								
DP Address: (in the server)		Object Type:	1254	Property ID:		111		
		Start-Index:	1	N° of elements		1		
Property access:		Read only ☐	-	Read/Write ☒				
Protection		Read level	-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐								
Special Features:								
None.								

2.2.6.3.2 FailSafeCurrent

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐	
Description:								
Maximum current that the station may draw when it is in failsafe mode								
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Float	Maximum station current when in failsafe mode		M	0-10.000	A	--	none	
Communication:								
DP Address: (in the server)		Object Type:	1254	Property ID:		112		
		Start-Index:	1	N° of elements		1		
Property access:		Read only ☐	-	Read/Write ☒				
Protection		Read level	-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐								
Special Features:								
None.								

2.2.6.3.3 FailSafeTimeout

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
The time value used to detect a failure. If no value on the inputs is received during this time, the unit goes in failsafe Mode: Failsafe values will be activated instead of the input values and the failsafe feedback output will be set.							

DPT:	Name	DPT_TimePeriodSec	DPT_ID	7.005	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Time Period	Parameter setting the time value used to detect a failure		M	0-1200	s	1 s	none
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:		113	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☐	-	Read/Write ☒			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐							
Special Features:							
If value is set to 0= no Detection of FailSafe is required							

2.2.6.4 Diagnostic Data: EVSE AC Station

2.2.6.4.1 ActivePowerMax

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
maximum power of the charging point that can be supplied to the vehicle							
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Float	Diagnostic value representing the max power of the charging point		M	0-10.000.000	W	--	none
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:		130	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Should always be of a higher value than MinTotalActivePower							
Special Features:							
None.							

2.2.6.4.2 ActivePowerMin

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
minimum power of the charging point that can be supplied to the vehicle							
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Float	Diagnostic value representing the minimum power of the charging point		M	0-10.000.000	W	--	none
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:		131	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Should always be of a lower value than MaxTotalActivePower							
Special Features:							
None.							

2.2.6.4.3 CurrentRated

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
Rated current of the charging point							
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Float	Diagnostic value representing the rated current of charging point		M	0-10.000	A	--	none
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:	132		
		Start-Index:	1	N° of elements	1		
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level	-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

2.2.6.4.4 CurrentMax

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
maximum current of the charging point that can be supplied to the vehicle							
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:		
Field	Description		Sup.	Range	Unit	Resol.:	Default
Float	Diagnostic value representing the max current of the charging point		M	0-10.000	A	--	none
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:	133		
		Start-Index:	1	N° of elements	1		
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level	-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Should always be of higher value then MinCurrent							
Special Features:							
None.							

2.2.6.4.5 CurrentMin

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐	
Description:								
minimum current of the charging point that can be supplied to the vehicle								
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:			
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Float	Diagnostic value representing the min current of the charging point		M	0-10.000	A	--	none	
Communication:								
DP Address: (in the server)		Object Type:	1254	Property ID:		134		
		Start-Index:	1	N° of elements		1		
Property access:		Read only ☒	-	Read/Write ☐				
Protection		Read level	-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Should always be of a lower value than MaxCurrent								
Special Features:								
None.								

2.2.7 Inputs, Outputs, Parameters and Diagnostic Data of EVSE AC Connector

2.2.7.1 Inputs: EVSE AC Connector

2.2.7.1.1 CurrentLimit

DP Name:	CurrentLimit	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Input for the upper limit of the current of the charging point that can be supplied to the vehicle. 0 Ampere pauses the charging session.					
Datapoint Type					
DPT Name:	DPT_Value_Electric_Current				
DPT Format:		DPT ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Setting the upper limit of the current of charging point	M	0-10.000	A	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	60	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Values below the minimal charging current of the station will automatically set the value to 0 Ampere. Values above the maximal charging current will set the value to Max Ampere. For FailSafe, see FailSafeTimeout					
Special Features					

2.2.7.1.2 ActivePowerLimit

DP Name:	ActivePowerLimit	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Input for the upper limit of the power of the charging point that can be supplied to the vehicle. 0 kW pauses the charging session.					
Datapoint Type					
DPT Name:	DPT_Value_Power				
DPT Format:		DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Setting the upper limit of the power of the charging point	M	0-10.000.000	W	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	61
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Values below the minimal charging power of the station will automatically set the value to 0 kW. Values above the maximal charging power will set the value to Max Power. For FailSafe, see FailsafeTimeout					
Special Features					

2.2.7.2 Outputs: EVSE AC Connector

2.2.7.2.1 CurrentLimitEff

DP Name:	CurrentLimitEff	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Feedback for the upper limit of the current of the charging point that can be supplied to the vehicle					
Datapoint Type					
DPT Name:	DPT Value Electric Current				
DPT Format:		DPT ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value reflecting the upper limit of the current of the charging point	M	0-10.000	A	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous Request	☒	Cyclically:	☐	Time-out:	
	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID): Start-Index:	1254 1	Property ID: N° of elements	90 1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.2 FailSafeCurrentEff

DP Name:	FailSafeCurrentEff	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Feedback of the current used during failsafe mode in a connector					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Current				
DPT_Format:		DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value reflecting the current used during failsafe mode		0-10.000A	A	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	91	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.7.2.3 ActivePowerLimitEff

DP Name:	ActivePowerLimitEff	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Feedback for the upper limit of the power of the charging point that can be supplied to the vehicle					
Datapoint Type					
DPT Name:	DPT_Value_Power				
DPT Format:		DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value reflecting the upper limit of power of the charging point	M	0-10.000.000	W	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	92
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.4 FailSafeActivePowerEff

DP Name:	FailSafeActivePowerEff	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Feedback of the power used during failsafe mode in station					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value representing the power using during failsafe mode	M	0-10.000.000	W	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	93
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.6 ActivePower

DP Name:	ActivePower	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present power of the charging point					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value representing the present charging point power	M	0-10.000.000	W	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
	Default Group Address:	none			
◆ Property				Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	94
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.7 CurrentL1

DP Name:	CurrentL1	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present Current of the charging point on phase 1					
Datapoint Type					
DPT_Name:	DPT_Value_Electric_Current				
DPT_Format:		DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing the present L1 current of the charging point	M	0-10.000	A	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	95
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.8 CurrentL2

DP Name:	CurrentL2	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present Current of the charging point on phase 2					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing the present L2 current of the charging point	M	0-10.000	A	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	96
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.9 CurrentL3

DP Name:	CurrentL3	Abbr.:		Mandatory	☒
FB Name:	EVSE			Can be internal	☐
Description					
Present Current of the charging point on phase 3					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Value_Electric_Current	DPT_ID:	14.019		
Field		Supp.	Range	Unit	Default
Float	Value representing the present L1 current of the charging point	M	0-10.000	A	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory:
Default Group Address:					none
◆ Property					Mandatory:
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	97
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.11 Control Pilot State

DP Name:	ControlPilotState	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Enum for different states of the charging point					
Datapoint Type					
DPT_Name:					
DPT Format:	20.1219	DPT_ID:	DPT_EVSEMode_1		
Field		Supp.	Range	Unit	Default
Field1	0 = Standby 1 = Vehicle Detected 2 = Ready charging 3 = Charging with ventilation 4 = Finished 5 = Reserved for specific EV 6 = unavailable 7 = ERROR 8 = Suspended_EV_SE (due to some limitations of the EVSE) 9 = Suspended_EV (due to signals from the EV) 10-255: reserved	M	Full	none	none
Access Type					
◆ Input					
N → this		☒	1 → this		☐
Spontaneous		☒	Cyclically:	☐	Time-out:
Request		☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	98	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.12 OCPP State

DP Name:	OCPPState		Abbr.:		Mandatory	☐
FB Name:	EVSE				Can be internal	☐
Description						
States of the Charging Point as an enum according to Open Charge Point Protocol (OCPP).						
Datapoint Type						
DPT_Name:						
DPT Format:	20.1220		DPT_ID:	DPT_EVSEMode_2		
Field		Supp.	Range	Unit	Default	
Field1	Value representing the states of the charging point according OCPP 0 = Charging 1 = EVConnected 2 = SuspendedEV 3 = SuspendedEVSE 4 = Idle 5 = Available 6 = Occupied 7 = Reserved 8 = Unavailable 9 = Faulted 10-255 = reserved	M	full	none	none	
Access Type						
◆ Input						
N → this	☒	1 → this	☐			
Spontaneous	☒	Cyclically:	☐	Time-out:		
Request	☐	Polling:	☐	Period:		
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		none				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	99		
	Start-Index:	1	N° of elements	1		
Property access:	Read only	☐	Read/Write	☒		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						
The Datapoint can be used for OCPP Stations with the software Version 1.6 or newer.						
Note: Preparing and Finishing states deriving from OCPP 1.6 are mapped to the new 2.0.1 state Occupied New States like EV Connected and Idle can be left untouched if the charging stations software is OCPP 1.6						

2.2.7.2.13 ISO15118 Detected

DP Name:	ISO15118 Detected		Abbr.:		Mandatory	☒
FB Name:	EVSE				Can be internal	☐
Description						
High level (digital) communication with the car is possible						
Datapoint Type						
DPT_Name:						
DPT Format:	DPT_Bool		DPT_ID:	1.002		
Field			Supp.	Range	Unit	Default
b	Value representing whether (1) or not (0) communication with the car is possible		M	full	bool	none
Access Type						
◆ Input						
	N → this	☒	1 → this	☐		
	Spontaneous	☒	Cyclically:	☐	Time-out:	
	Request	☐	Polling:	☐	Period:	
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:		none			
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	512	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☐	Read/Write	☒	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

2.2.7.2.14 StateOfCharge

DP Name:	StateOfCharge	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
State of charge of electric vehicle in %					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Scaling	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value representing the state of charge	M	full	%	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
	Default Group Address:	none			
◆ Property				Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	513
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.15 BatteryCapacity

DP Name:	BatteryCapacity	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Battery capacity of electric vehicle in Watt					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value representing the battery capacity	M	0 – 1.000.000	Wh	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
	Default Group Address:	none			
◆ Property				Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	514
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.16 DepartureTime

DP Name:	DepartureTime	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Departure time of electric vehicle in seconds since start of charging session					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_TimePeriodSec	DPT_ID:	7.005		
Field		Supp.	Range	Unit	Default
TimePeriod	Value representing departure time of vehicle in seconds since start of charging	M	full	s	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	515	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.17 StateOfChargeMin

DP Name:	StateOfChargeMin	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Minimum State of Charge EV needs to keep throughout the charging session					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Scaling	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value representing the minimum state of charge the EV needs to keep during charging	M	full	%	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	516
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.2.18 StateOfChargeTarget

DP Name:	StateOfChargeTarget	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
Target SoC for departure					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Scaling	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value representing the target state of charge for departure	M	full	%	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	517	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

2.2.7.2.19 StateOfChargeTargetEnergyReq

DP Name:	StateOfChargeTargetEnergyReq		Abbr.:		Mandatory	☐
FB Name:	EVSE				Can be internal	☐
Description						
Energy required to reach target SoC for departure						
Datapoint Type						
DPT_Name:						
DPT Format:	DPT_ActiveEnergy		DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default	
Signed Value	Value representing the energy required to reach target state of charge for departure		M	0 – 1.000.000	Wh	none
Access Type						
◆ Input						
	N → this	☒	1 → this	☐		
	Spontaneous	☒	Cyclically:	☐	Time-out:	
	Request	☐	Polling:	☐	Period:	
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	1254	Property ID:	518	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☐	Read/Write	☒	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

2.2.7.2.20 RFIDTag

DP Name:	RFIDTag	Abbr.:		Mandatory	☐
FB Name:	EVSE			Can be internal	☐
Description					
RFID of the plugged in EV					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_VarString_8859_1	DPT_ID:	24.001		
Field		Supp.	Range	Unit	Default
String	Value reflecting the RFID of the plugged EV	M	full	none	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	1254	Property ID:	519	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

2.2.7.3 Parameter: EVSE AC Connector

2.2.7.3.1 FailSafeActivePower

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
Maximum power that the station may draw when it is in failsafe mode					
DPT:	Name	DPT Value Power	DPT_ID	14.056	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Parameter setting the max power a station may draw in safe mode		M	0 – 10.000.000	W
					Resol.:

					Default
					none
Communication:					
DP Address: (in the server)		Object Type:	1254	Property ID:	120
		Start-Index:	1	N° of elements	1
Property access:		Read only ☐	-	Read/Write ☒	
Protection		Read level	-	Write level	-
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐					
Special Features:					
None.					

2.2.7.3.2 FailSafeCurrent

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
Maximum current that the station may draw when it is in failsafe mode					
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Parameter setting the max current a station may draw in safe mode		M	0- 10.000	A
					Resol.:
					--
					Default
					none
Communication:					
DP Address: (in the server)		Object Type:	1254	Property ID:	121
		Start-Index:	1	N° of elements	1
Property access:		Read only ☐	-	Read/Write ☒	
Protection		Read level	-	Write level	-
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐					
Special Features:					
None.					

2.2.7.4 Diagnostic Data: EVSE AC Connector

2.2.7.4.1 ActivePowerMax

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
maximum power of the charging point that can be supplied to the vehicle							
DPT:	Name	DPT Value	Power	DPT_ID	14.056	Datatype format:	
Field	Description	Sup.	Range	Unit	Resol.:	Default	
Float	Diagnostic data representing the max power of the charging point	M	0-10.000.000	W	--	none	
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:		140	
		Start-Index:	1	N° of elements		1	
Property access:		Read only	☒ -	Read/Write		☐	
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Should always be of a higher value than Connector Min Total Active Power							
Special Features:							
None.							

2.2.7.4.2 ActivePowerMin

FB:	EVSE	Property Name (Server):		Mandatory	☒	Optional	☐
Description:							
minimum power of the charging point that can be supplied to the vehicle							
DPT:	Name	DPT Value	Power	DPT_ID	14.056	Datatype format:	
Field	Description	Sup.	Range	Unit	Resol.:	Default	
Float	Diagnostic data representing the min power of the charging point	M	0-10.000.000	W	--	none	
Communication:							
DP Address: (in the server)		Object Type:	1254	Property ID:		141	
		Start-Index:	1	N° of elements		1	
Property access:		Read only	☒ -	Read/Write		☐	
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Should always be of lower value then Max Total Active Power							
Special Features:							
None.							

2.2.7.4.3 CurrentMax

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
maximum current of the charging point that can be supplied to the vehicle					
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Diagnostic data representing the max current of the charging point		M	0-10.000	A
Communication:					
DP Address: (in the server)		Object Type: 1254	Property ID: 142		
		Start-Index: 1	N° of elements 1		
Property access:		Read only ☒ -	Read/Write ☐		
Protection		Read level -	Write level -		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
Should always be of a higher value than Connector Min Current					
Special Features:					
None.					

2.2.7.4.4 CurrentMin

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
minimum current of the charging point that can be supplied to the vehicle					
DPT:	Name	DPT_Value_Electric_Current	DPT_ID	14.019	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Diagnostic data representing the min current of the charging point		M	0-10.000	A
Communication:					
DP Address: (in the server)		Object Type: 1254	Property ID: 143		
		Start-Index: 1	N° of elements 1		
Property access:		Read only ☒ -	Read/Write ☐		
Protection		Read level -	Write level -		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
Should always be of lower value then Connector Max Current					
Special Features:					
None.					

2.2.7.4.5 Bidirectional EV Detected

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
Support for bidirectional charging by the charging station/EV					
DPT:	Name	DPT_Bool	DPT_ID	1.002	Datatype format:
Field	Description		Sup.	Range	Unit
B	Diagnostic data informing whether bidirectional charging is supported (1) or not (0)		M	full	bool
					Resol.:
					--
					Default
					none
Communication:					
DP Address:		Object Type:	1254	Property ID:	144
(in the server)		Start-Index:	1	N° of elements	1
Property access:		Read only ☒	-	Read/Write ☐	
Protection		Read level	-	Write level	-
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
Special Features:					
None.					

2.2.7.4.6 DischargePowerMax

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
Max possible power that can be drawn from the EV through bidirectional charging					
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Diagnostic data informing about the max possible power that can be drawn from EV through bidirectional charging		M	0-10.000.000	W
					Resol.:
					--
					Default
					none
Communication:					
DP Address:		Object Type:	1254	Property ID:	150
(in the server)		Start-Index:	1	N° of elements	1
Property access:		Read only ☒	-	Read/Write ☐	
Protection		Read level	-	Write level	-
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
Special Features:					
None.					

2.2.7.4.7 DischargePowerMin

FB:	EVSE	Property Name (Server):		Mandatory	☒
				Optional	☐
Description:					
Min possible power that can be drawn from the EV through bidirectional charging					
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:
Field	Description		Sup.	Range	Unit
Float	Diagnostic data informing about the min possible power that can be drawn from the EV through bidirectional charging		M	0-10.000.000	W
					Resol.:
					--
					Default
					none
Communication:					
DP Address:		Object Type:	1254	Property ID:	151
(in the server)		Start-Index:	1	N° of elements	1
Property access:		Read only ☒	-	Read/Write ☐	
Protection		Read level	-	Write level	-
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐					
Special Features:					
None.					

2.2.8 New Data Point Types

2.2.8.1 Control Pilot State

<u>Format:</u>	1 octet: N_8												
octet nr.	1												
field names	<table border="1"><tr><td>EVSE Mode</td></tr></table>					EVSE Mode							
EVSE Mode													
encoding	<table border="1"><tr><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td></tr></table>					N	N	N	N	N	N	N	N
N	N	N	N	N	N	N	N						
<u>Range:</u>													
<u>Unit:</u>													
<u>Resol.:</u>	none												
<u>PDT:</u>													
Datapoint Types													
<u>ID:</u>	<u>Name:</u>	<u>Range:</u>	<u>Unit:</u>	<u>Resol.:</u>	<u>Use:</u>								
20.1219	DPT_EVSEMode_1	See below	See below	See below	EVSE								
Data fields		Description	Unit/Range										
EVSEMode		EVSE Operating mode	0 = Standby (available for new EV) 1 = Vehicle Detected (new Customer detected, preparing of charging session) 2 = Ready charging (preparation complete, waiting for plug in) 3 = Charging with ventilation 4 = Finished (Charging process finished, e.g. vehicle fully charged or process aborted) 5 = Reserved (Charging point is reserved for specific EV, no other EV is allowed in particular timeframe) 6 = unavailable (charging station is currently unavailable) 7 = ERROR (the Charging Station is malfunctioning and needs repairing or servicing.) 8 = Suspended_EV_SE (Termination of the charging process due to some limitations of the EVSE) 9 = Suspended_EV (Termination of the charge process due to signals from the EV) 10-255: reserved										

2.2.8.2 OCPP State

<u>Format:</u>	1 octet: N_8												
octet nr.	1												
field names	<table border="1"><tr><td>EVSE Mode</td></tr></table>					EVSE Mode							
EVSE Mode													
encoding	<table border="1"><tr><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td></tr></table>					N	N	N	N	N	N	N	N
N	N	N	N	N	N	N	N						
<u>Range:</u>													
<u>Unit:</u>													
<u>Resol.:</u>	none												
<u>PDT:</u>													
Datapoint Types													
<u>ID:</u>	<u>Name:</u>	<u>Range:</u>	<u>Unit:</u>	<u>Resol.:</u>	<u>Use:</u>								
20.1220	DPT_EVSEMode_2	See below	See below	See below	EVSE								
Data fields		Description	Unit/Range										
EVSEMode		EVSE Operating mode	States of the Charging Point according to OCPP 0 - Charging 1 - EVConnected 2 - SuspendedEV 3 - SuspendedEVSE 4 - Idle 5 - Available 6 - Occupied 7 - Reserved 8 – Unavailable 9 - Faulted 10-255: reserved										

2.2.8.3 EVSEErrorCode

<u>Format:</u>	1 octet: N_8												
octet nr.	1												
field names	<table border="1"><tr><td>EVSE Mode</td></tr></table>					EVSE Mode							
EVSE Mode													
encoding	<table border="1"><tr><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td></tr></table>					N	N	N	N	N	N	N	N
N	N	N	N	N	N	N	N						
<u>Range:</u>													
<u>Unit:</u>													
<u>Resol.:</u>	none												
<u>PDT:</u>													
Datapoint Types													
<u>ID:</u>	<u>Name:</u>	<u>Range:</u>	<u>Unit:</u>	<u>Resol.:</u>	<u>Use:</u>								
20.1221	DPT_EVSEMode_3	See below	See below	See below	EVSE								
Data fields	Description			Unit/Range									
EVSEMode	error code of the charging point			1. Connector Lock Failure - Failure to lock or unlock connector 2. EV Communication Error - Communication failure with the vehicle, might be Mode 3 or other communication protocol problem. This is not a real error in the sense that the Charge point doesn't need to go to the faulted state. Instead, it should go to the SuspendedEVSE state 3. Ground Failure - Ground fault circuit interrupter has been activated 4. High Temperature - Temperature inside Charge point is too high 5. internal Error - Error in internal hard- or software component 6. Local List Conflict - The authorization information received from the central System is in conflict with the Local Authorization List 7. NoError - No error to report 8-255: reserved									